Exploratory Data Analysis of Netflix Catalog Trends: Geographical, Demographic, and Quality Insights

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***Abstract*—The rapid expansion of global streaming platforms has fundamentally altered media consumption, yet understanding the underlying content strategies requires robust data analysis. This study presents an exploratory data analysis of the Netflix content catalog to identify key trends in production geography, target demographics (via age ratings), and historical content quality. Using a dataset of Netflix titles and credits, we applied Python-based data cleaning techniques to handle missing values, impute unrated data, and parse stringified arrays. Our visual analysis reveals that a highly concentrated group of countries dominates production share, while mature ratings (TV-MA and R) make up a significant portion of both movies and TV shows. Furthermore, mapping historical IMDb scores from 1970 to 2022 illustrates a noticeable shift in average rating trajectories during Netflix's aggressive catalog expansion phase (2015–2021). These insights provide a quantitative overview of streaming content acquisition and production strategies.**

Keywords—Exploratory Data Analysis, Python, Pandas, Data Visualization, Streaming Media, Content Strategy.

I. INTRODUCTION

The rise of digital streaming platforms has transformed the entertainment industry, with platforms like Netflix shifting from licensed content distributors to major global production studios [1]. As these platforms scale globally, understanding their catalog composition is critical for media analysts, rival networks, and consumers affected by these shifting media landscapes.

Despite the availability of vast amounts of catalog data, raw datasets often contain inconsistencies, such as stringified lists and missing quantitative values, making immediate analysis difficult. This study addresses this practical problem by standardizing a messy dataset and extracting clear, visual insights regarding Netflix's catalog strategy. The objective of this study is to clean and analyze this data to highlight geographical production concentrations, compare age demographics between movies and TV shows, and track quality metrics (IMDb scores) over a 50-year period.

The main contribution of this paper is a reproducible data cleaning and visualization pipeline that highlights the impacts of Netflix's aggressive expansion on content quality. The remainder of this paper details the methodology for dataset preparation, presents the visualizations, and discusses the implications of the findings.

II. RELATED WORK

Previous studies and independent data analyses on streaming ecosystems frequently focus on recommendation algorithms or subscriber growth models. While existing approaches have mapped Netflix's basic genre distributions, many analyses fail to adequately clean complex nested data structures (such as arrays imported as strings) or address the high volume of missing values in runtimes and age certifications. This work addresses that specific gap by implementing robust preprocessing using Python's Abstract Syntax Tree (`ast`) module before visualizing the data [2].

III. METHODOLOGY

A. Study Design and Materials

This study utilizes an observational design based on a publicly available Netflix dataset [1] consisting of two primary files: `titles.csv` and `credits.csv`. The dataset contains structural metadata (type, release year, age certification, runtime, genres) and quality metrics (IMDb scores).

The software environment used for this analysis was Python, utilizing the Pandas [2] and NumPy libraries for data manipulation, and Matplotlib and Seaborn for generating visualizations.

B. Procedure and Evaluation

To ensure accurate analysis and allow for reproducibility, the data underwent a strict preprocessing pipeline:

Filtering: Titles lacking an IMDb score were immediately dropped from the main dataframe.

Parsing Complex Strings: Raw data stored lists (like genres and production countries) as text strings (e.g., "['Comedy']"). We defined a custom function utilizing `ast.literal_eval` to safely convert these back into iterable Python lists, falling back to string manipulation for corrupted entries.

Data Imputation: Movies inherently lack television seasons, so missing seasons values were filled with 0 and cast to integers. Missing `age_certification` values were explicitly labeled as "Unrated" to prevent row loss during demographic analysis. Missing runtime values were imputed using the dataset's median runtime to prevent skewing distribution metrics.

Following the cleaning phase, Seaborn was used to evaluate production volume, runtime distributions, catalog share by country, age certification by content type, historical IMDb score averages, metric correlations, top talent, and a strategic quality-volume portfolio view.

IV. RESULTS AND DISCUSSION

Table I summarizes the cleaning transformations applied before analysis. The cleaned data was then used to produce

nine visual insights into the structure of the Netflix catalog (Figs. 1–9).

TABLE I
IMPACT OF DATA CLEANING ON DATASET

Feature/Metric	Raw Dataset State	Proposed Cleaned State
genres / countries	Text strings (e.g., "['US']")	Iterable Python lists
runtime (missing)	NaN values	Filled with column median
seasons (movies)	NaN values	Forced to integer 0
age_certification	NaN values	Labeled as "Unrated"

A. Movies vs. TV Shows Production Volume (1945–2022) (Fig. 1)

Across more than seven decades, the most striking trend is the explosion of content added to the platform starting around 2015. This marks the turning point where Netflix shifted from licensing older titles to aggressively producing original content. Both formats grew, but feature films dominated the expansion, significantly outpacing new TV show releases.

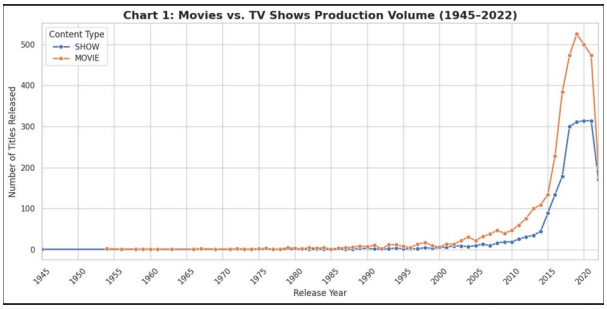


Fig. 1. Movies vs. TV shows production volume (1945–2022).

B. Runtime Distribution and Outlier Detection (Fig. 2)

A clear split emerges in how the two formats are structured. Movies are highly standardized, with most titles clustering around a 98 to 102-minute median runtime. TV shows show a much tighter baseline for standard episode lengths but are heavily skewed by extreme outliers, which reflect investment in limited mini-series and extended special episodes that break traditional network time constraints.

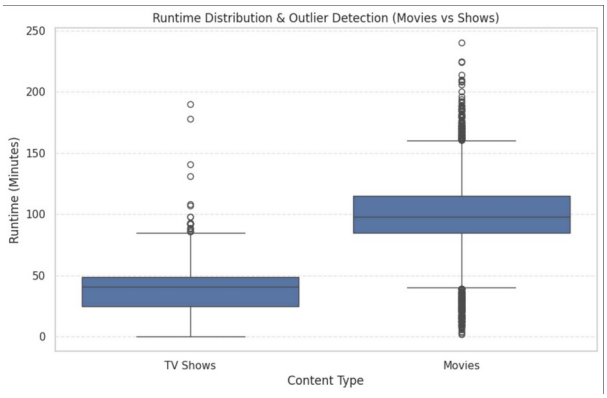


Fig. 2. Runtime distribution and outlier detection for movies vs. shows.

C. Top 10 Producing Countries by Catalog Share (Fig. 3)

The production footprint is remarkably concentrated. The United States and India stand out as the dominant markets, supplying the vast majority of the standalone catalog. Beyond these two, Netflix relies on established creative hubs in Europe and East Asia, specifically the United Kingdom, Japan, and South Korea, as regional pillars.

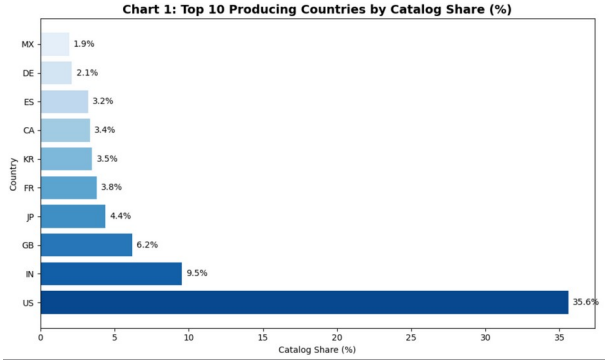


Fig. 3. Top 10 content producing countries by catalog share.

D. Age Certification and Content Maturity (Fig. 4)

The catalog heavily targets older audiences rather than families or young children. TV shows lean overwhelmingly toward the mature TV-MA rating, while feature films follow a similar pattern, favoring adult-oriented R and TV-14 ratings. This underscores a deliberate strategy of focusing on mature content over all-ages programming.

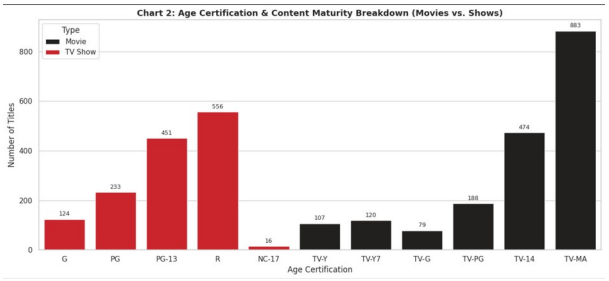


Fig. 4. Age certification breakdown for movies vs. shows.

E. 50-Year Historical Quality Trajectory (1970–2022) (Fig. 5)

Episodic TV shows weathered the content boom well, maintaining a consistently high and stable average IMDb rating between 6.8 and 7.2 over the decades. Movies, however, suffered a noticeable "volume dilution" effect: as Netflix ramped up its feature film catalog between 2015 and 2021, the average movie rating drifted downward, suggesting the rapid increase in production volume coincided with slightly lower viewer ratings.

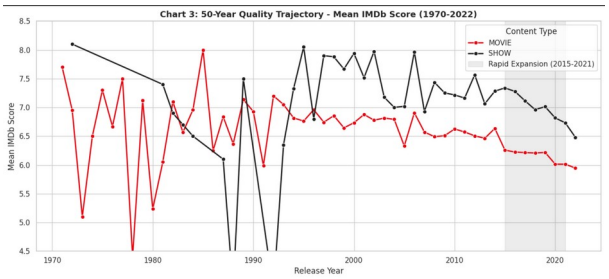


Fig. 5. Mean IMDb scores by release year (1970–2022).

F. Correlation Heatmap Matrix (Fig. 6)

The total number of viewer votes (`imdb_votes`) acts as a much stronger indicator of a title's quality score than raw popularity metrics. This suggests that deep audience engagement, where viewers care enough to rate a title, is a better validator of quality than simple everyday popularity.

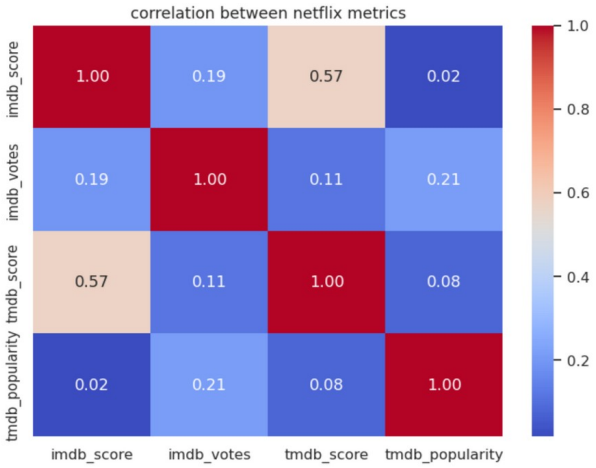


Fig. 6. Correlation heatmap of numerical features.

G. IMDb Score vs. TMDB Popularity (Fig. 7)

Comparing critical rating (`imdb_score`) against platform popularity (`tmdb_popularity`) shows a surprisingly weak relationship, with a Pearson correlation coefficient of about 0.22. Going viral and achieving massive streaming volumes does not necessarily mean a title is a high-quality, critically acclaimed piece of media.

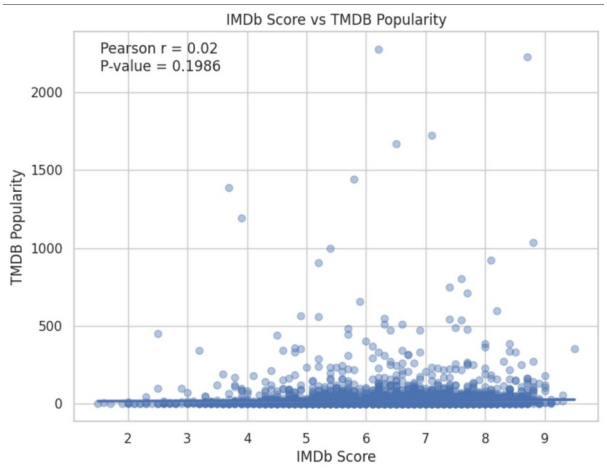


Fig. 7. IMDb score vs. TMDB popularity scatter and regression plot.

H. Top 10 Directors and Actors Leaderboard (Fig. 8)

Filtering to creators with a proven track record of three or more projects on Netflix yields a clear leaderboard of top talent. This helps identify which recurring directors and actors bring the most consistent value and critical success to the platform.

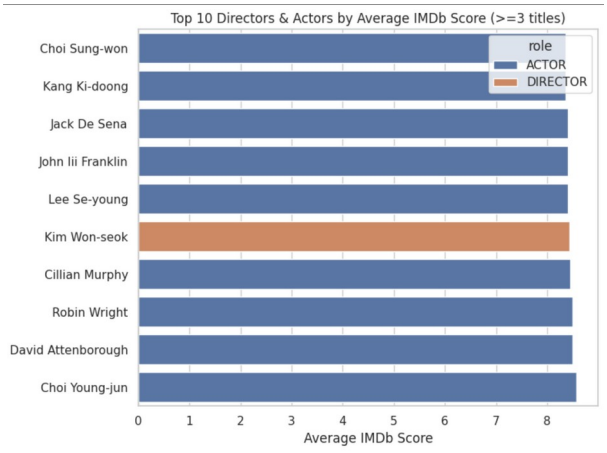


Fig. 8. Top 10 directors and actors leaderboard.

I. Strategic Portfolio Quadrant Analysis (Fig. 9)

This bubble plot maps the portfolio along three factors: volume (how much content exists), quality (how good it is), and engagement depth (bubble size). Dividing the catalog into four quadrants separates high-volume "filler" titles that keep users occupied from high-quality "prestige" anchors designed to win awards and attract new subscribers.

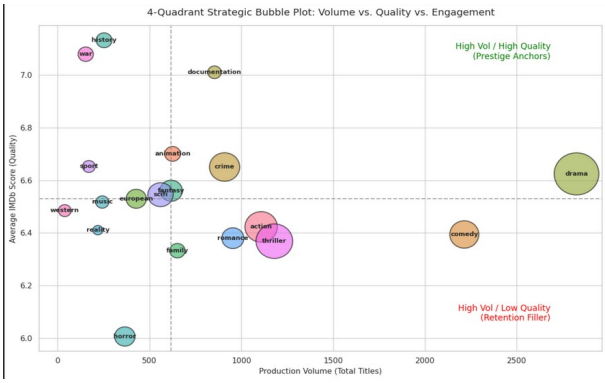


Fig. 9. Four-quadrant bubble plot: volume vs. quality vs. engagement depth.

V. CONCLUSION

This analysis successfully mapped key aspects of the Netflix catalog, revealing that the platform relies heavily on US-centric production and mature-rated content. The historical IMDb data also suggests that the platform's strategic pivot toward high-volume original content creation between 2015 and 2021 coincided with a slight dip in average viewer ratings. A key limitation of this study is the reliance on IMDb scores, which reflect subjective audience sentiment rather than objective viewership metrics. Future work should integrate actual viewership hours or engagement data to determine if lower-rated content during the expansion phase still achieved commercial success.

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